MIL-D-52718A(ME)
2 October 1979
SUPERSEDING
MIL-D-52718(ME)
24 September 1971

MILITARY SPECIFICATION

DEMINERALIZER, WATER, ION EXCHANGE UNIT

AND

DEMINERALIZATION EQUIPMENT SET, WATER, ION EXCHANGE

3,000 GPH; PACKAGING OF

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for materials, ıethods, and procedures for the preservation, packing, and marking of the ı,000 GPH demineralizer unit and demineralization equipment set.

2. APPLICABLE DOCUMENTS

2.1 Issue of document. The following documents of the issue in effect ın date of invitation for bids or request for proposal form a part of his specification to the extent specified herein:

PECIFICATIONS

FEDERAL

FF-N-105 -Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.

FSC-PACK

eneficial comments (recommendations, additions, deletions) and any pertinent ata which may be of use in improving this document should be addressed to: S Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, ort Belvoir, VA 22060 by using the self-addressed Standardization Document mprovement Proposal (DD Form 1426) appearing at the end of this document r by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

NN-P-530	-Plywood, Flat-Panel.
QQ-S-781	-Strapping, Steel and Seals.
PPP-B-601	-Boxes, Wood, Cleated-Plywood.
PPP-B-636	-Box, Shipping, Fiberboard.
PPP-P-40	-Packaging and Packing of Hand Tools.
PPP-T-60	-Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	-Preservation - Packaging, Methods of.
MIL-B-121	-Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-10603	-Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-T-14398	-Tanks, Fabric, Collapsible: Water, Nylon, Rubber-Coated.
MIL-G-28554	-Generator Sets, Mobile Electric Power and Supplemental Equipment; Packaging of.

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-281	-Automobiles, Trucks, Truck-Tractors, Trailers and Trailer-Dollies; Preservation and Packaging of.
MIL-STD-731	-Quality of Wood Members for Containers and Pallets.
MIL-STD-1188	-Commerical Packaging of Supplies and Equipment.

DRAWINGS

ME

13217E9357	-Cover, Pumps, Cotton Duck, Insulated.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this pecification to the extent specified herein. Unless otherwise indicated, he issue in effect on date of invitation for bids or request for proposal hall apply.

DEPARTMENT OF COMMERCE

PS-1 - Construction and Industrial Plywood.
PS-51 - Hardwood and Decorative Plywood.

Application for copies should be addressed to the Superintendent f Documents, US Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Preproduction pack. The contractor shall furnish a preproduction ack for examination within the time frame specified (see 6.2) to prove, rior to starting production packaging that the applied preservation, packng and marking comply with the requirements of this specification. xamination shall be as specified in section 4 and shall be subject to urveillance and approval by the Government (see 6.3). The preproduction ack may be prepared utilizing either the preproduction model or a production demineralizer. When the preproduction model is utilized, any preseration and packing shall be removed by the contractor at no expense to the overnment, when requested by the Government, to facilitate comparison etween the preproduction model and production demineralizer.

3.2 Materials. Materials shall be as specified herein and in the applible specifications. Materials not specified shall be selected by the ontractor and shall be subject to all provisions of this specification see 6.5).

3.3 Preservation. Preservation shall be level A or Commercial as pecified (see 6.2). Consolidation requirements specified herein have oven adequate to contain all accessories and components. However, when ie configuration of these items result in dimensional changes, the conactor may relocate such items in any available space provided equivalent otection is afforded the item. Relocation of any item shall be subject approval of the contracting officer.

3.3.1 Level A

3.3.1.1 Preservatives. Preservatives specified herein shall conform to ie applicable specifications listed in and shall be applied in accordance ith MIL-P-116. CAUTION: Contact preservatives shall not be applied to ıy surface, other than the interior of the pump, which will come in ntact with chemical solutions, slurrier, or water, except waste water.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the current version before use.

3.3.1.2 Demineralizer, unit.

3.3.1.2.1 Cable assemblies. The unconnected cable assemblies shall be coiled to a minimum coil not less than 10 times the cable diameter. The coils shall be secured with equally spaced wraps of cotton tape or twine.

3.3.1.2.2 Strap assemblies. Strap assemblies not used for tiedown of components shall be rolled to the minimum size roll. The roll shall be secured with cotton tape or twine.

3.3.1.2.3 Centrifugal pump. The centrifugal pump shall be preserved in accordance with MIL-P-10603, level A, except where P-10 preservative is specified, P-14 preservative shall be used.

3.3.1.2.4 Pump cover. The pump cover shall be preserved in accordance with Drawing 13217E9357, level A.

3.3.1.2.5 Hose assembly, (garden type). The hose assembly shall be coiled in a minimum size coil and the coil secured with cotton tape or twine.

3.3.1.2.6 Ground rods, with clamp and wire. The ground wire shall be coiled and secured to the ground rod with tape conforming to PPP-T-60, type IV.

3.3.1.2.7 Fuel pump cable assembly and fuel lines. The fuel pump cable assembly and fuel line shall be coiled together in a minimum size coil and the coil secured with cotton tape or twine.

3.3.1.2.8 Drum restrainer. The drum restrainer clamps on each end of the restrainer shall be secured to the frame of the restrainer with tape conforming to PPP-T-60, type IV.

3.3.1.2.9 Acid hose assembly. The acid hose assembly shall be looped or coiled to fit the space below the demineralizer tanks (see figure 1). The loop or coil shall be secured with cotton tape or twine.

3.3.1.2.10 Fabric enclosure assembly. The fabric enclosure assembly shall be folded to a dimension compatible with the space provided for the enclosure directly behind the demineralizer tanks (see figure 1).

3.3.1.2.11 Heater. All fuel shall be completely drained from the heater.

3.3.1.2.12 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.3.1.2.13 Maintenance tools. Maintenance tools shall be preserved
accordance with PPP-P-40, level A.

3.3.1.2.14 Technical publications. Technical publications shall be
eserved in accordance with MIL-P-116, method IC-1.

3.3.1.2.15 Piping valves. Except for valves DV20 and DV21, valves in
e piping system shall be left open.

3.3.1.2.16 Shelter windows . Each shelter window shall be covered with
4-inch plywood. The plywood shall be secured to the shelter with tape
nforming to PPP-T-60, type IV.

3.3.1.2.17 Consolidation. The fuel pump, fuel pump cable-fuel
ne assembly, and paddle shall be consolidated in the shelter
cessory chest in accordance with the chest loading plan. The 6 foot
d 50 foot cable assemblies and rolled strap assemblies shall be
nsolidated in the trailer accessory chest in accordance with the
cessory chest loading plan. The repair parts and maintenance tools
all be consolidated together in a close-fitting box conforming to
P-B-636, W5c, style optional. Technical publications shall be placed
the canvas case provided with the equipment.

3.3.1.2.18 Placement of components within the shelter. The shelter
d trailer accessory chests, drum restrainer, pump assembly, pump
sembly cover, garden type hose assembly, acid hose assembly, ground
ds and fabric enclosure assembly shall be placed and secured in the
elter in accordance with figure 1. The enclosure assembly support
ds shall be placed in the rack with the fabric enclosure assembly
shioning and blocking shall be provided between the trailer accessory
est and heater. The consolidated repair parts and maintenance tools
all be placed in the area behind the demineralizer tanks with the pump
ers. Maintenance cases containing the technical publications shall
secured to the parent equipment for which they are intended.

3.3.1.3 Demineralization equipment set.

3.3.1.3.1 Demineralizer units. The demineralizer unit shall be preserved,
nsolidated and components placed within the shelter as specified in 3.3.1.2.

3.3.1.3.2 Adapters. The external threads of each adapter shall be covered
:h a wrap of tape conforming to PPP-T-60, type IV. Note: Where adapters
e furnished in a contractors close-fitting container, taping of external
:eads shall not be required.

3.3.1.3.3 Power cable. The power cable shall be coiled in a minimum
.l not less than 10 times the cable diameter. The coil shall be secured
:h cotton tape or twine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.3.1.3.4 Pump covers. The pump covers shall be preserved in accordanc with Drawing 13217E9357, level A.

3.3.1.3.5 Gloves. Each pair of gloves shall be preserved in accordance with MIL-P-116, method IC-1.

3.3.1.3.6 Polyester fiber hoses. Each hose shall be rolled to a minimum size roll and the roll shall be secured with cotton tape of twine. Note: When hoses are furnished in vendor's close-fitting containe the container shall be retained as the required preservation.

3.3.1.3.7 Rubber hose assemblies, 1-1/4-inch ID and 1-1/2-inch ID . The couplings of each hose assembly shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The barrier shall be secured with tape conforming to PPP-T-60, type IV.

3.3.1.3.8 Rubber hose assemblies, 2-inch ID. Each hose assembly shall be bent to form a circle and the end couplings shall be connected tc secure the hose assembly in the circle pattern.

3.3.1.3.9 Extension light. The cable shall be coiled in a minimun coi not less than 10 times the cable diameter. The coil shall be secured witl cotton tape or twine.

3.3.1.3.10 Centrifugal pumps. The centrifugal pumps shall be preservec in accordance with MIL-P-10603, level A, except when P-10 preservative is specified P-14 preservative shall be used.

3.3.1.3.11 Collapsible fabric tanks. The 500 gallon and 3,000 gallon tanks shall be preserved in accordance with MIL-T-14398, level A.

3.3.1.3.12 Cargo trailer. The trailer components shall be preserved in accordance with MIL-STD-281, level A (mobile).

3.3.1.3.13 Generator set. The generator set shall be preserved in accordance with MIL-G-28554, level A.

3.3.1.3.14 Wrenches. Open end wrenches, pipe wrenches, spanner wrench and plug wrenches shall be preserved in accordance with PPP-P-40, level A.

3.3.1.3.15 Consolidation. The utility pails shall be consolidated in the trailer accessory chest along with the cable assemblies and strap assemblies not used for tiedown of equipment. The pails shall be nested within the cable assembly coils as indicated in the chest loading plan. The gate valves, wrenches, adapters, extension light, measure and gloves shall be consolidated in the shelter accessory chest along with the fuel pump, fuel pump cable-fuel line assembly, and paddle in accordance with chest loading plan.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the currentversion before use.

3.3.1.3.16 Placement of components within the shelter. The centrifugal pumps and pump covers shall be placed and secured in the shelter along with the demineralizer unit components in accordance with figure 1. Cushioning and blocking shall be provided between the trailer accessory chest and pumps and shelter accessory chest and pumps.

3.3.2 Commercial. Each demineralizer unit or demineralization equipment set shall be preserved in accordance with MIL-STD-1188.

3.4 Packing. Packing shall be level A or Commercial as specified (see 6.2).

3.4.1 Level A.

3.4.1.1. Demineralizer unit. The side-opening and door of the shelter shall be closed and secured. No additional packing of the unit is required.

3.4.1.2 Demineralization equipment set .

3.4.1.2.1 Power cable and hose assemblies. The 2-inch coupled hose assemblies shall be placed in a box with maximum inside dimensions of 45-inches long by 35-1/2-inches wide by 20-inches high conforming to PPP-B-601, overseas type, style I, except that strapping shall be located parallel to the skids and over end cleats. The hoses shall be placed so that the couplings of each layer will be on opposite sides of the box. The 1-1/4-inch and 1-1/2-inch hoses shall be coiled within the 2-inch hoses. The polyester fiber hoses and power cable shall then be placed within the coiled 1-1/4-inch and 1-1/2-inch hoses.

3.4.1.2.2 Calcium hypochlorite. The calcium hypochlorite shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style I, except that strapping shall be located parallel to the skids and over end cleats (see 6.4).

3.4.1.2.3 Hydrochloric acid. Hydrochloric acid shall be packed in quantities of six drums per box. The boxes shall be close-fitting and shall conform to PPP-B-601, overseas type, style I, except that strapping shall be located parallel to the skids and over end cleats (see 6.4).

3.4.1.2.4 Sodium carbonate. The 100 pound bags of sodium carbonate shall be palletized in accordance with figure 2.

3.4.1.2.5 Collapsible fabric tanks. The 500-gallon and 3,000-gallon tanks shall be packed in accordance with MIL-T-14398, level A.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

3.4.1.2.6 Cargo trailer. The cargo trailer, preserved as specified in 3.3.1.3.12 shall not require packing.

3.4.1.2.7 Generator set. The generator set shall be packed in accordance with MIL-G-28554, level A.

3.4.1.2.8 Demineralizer unit. The demineralizer unit shall be prepared as specified in 3.4.11.

3.4.2 Commercial. Each complete demineralizer unit or demineralization equipment set shall be packed in accordance with MIL-STD-1188.

3.5 Marking.

3.5.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

3.5.2 Commercial packaging . Marking shall be in accordance with MIL-STD-1188.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects as listed in 4.4.3. Presence of one or more defects shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the following defects. Acceptable quality level (AQL) shall be 2.5 percent defective.

101. Materials, methods or containers not as specified. Each incorrect material, method or container shall constitute one defect.

102. Cable assemblies not coiled and secured as specified.

103. Strapping assemblies not rolled and secured as specified.

104. P-14 preservative not used in lieu of P-10 preservative of centrifugal pumps.

105. Pump covers not preserved as specified.

106. Hose assemblies not rolled, coiled or bent, and secured as specified.

107. Ground rod wire not coiled and secured as specified.

108. Fuel pump cable assembly and fuel lines not coiled and secured as specified.

109. Drum restrainer clamps not secured as specified.

110. Acid hose assembly not looped or coiled and secured as specified.

111. Fabric enclosure not folded as specified.

112. Heater not completely drained of fuel as specified.

113. Repair parts not preserved as specified.

114. Maintenance tools not preserved as specified.

115. Technical publications not preserved as specified.

116. Piping valves not closed as specified.

117. Shelter windows not covered with plywood as specified.

118. Consolidations not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

119. Placement of components within the shelter not as specified.

120. Adapter threads not protected with tape as specified.

121. Gloves not preserved as specified.

122. Hose couplings not covered with barrier material as specified.

123. Cushioning and blocking not provided as specified.

124. Hydrochloric acid not packed in quantities specified.

125. Sodium carbonate not palletized in accordance with figure 2.

126. Marking illegible, incomplete, incorrect or missing.

5. PACKING

(This section not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts and in section 5 of commodity specifications for the preservation, packing and marking of demineralizer units and demineralizati equipment sets.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Time frame required for submission of a preproduction pack (see 3.1).

(c) Degree of preservation and degree of packing required (see 3.3 and 3.4).

(d) When fabrication of pallet cap will be other than as specified in figure 2. (See figure 2, Note 6).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the demineral units and demineralization equipment sets in accordance with this specifica

6.4 Boxes of the following inside dimensions have been found adequate fc packing of the following items:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

Items	Box Size
Calcium hypochlorite	-Length - 30-1/2 inches Width - 20-1/4 inches Height - 18-1/2 inches
Hydrochloric acid	-Length - 46-3/4 inches Width - 31-1/4 inches Height - 25-3/4 inches

Depending on contractor's product container size, dimensions of boxes are subject to change. All boxes shall be close-fitting to produce a pack of minimum cube.

6.5 Recycled material. It is encouraged that recycled material be used when practical, as long as it meets the requirements of this specification (See 3.2).

Custodian:
Army - ME

Preparing Activity:
Army - ME

Project No. PACK-A178

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

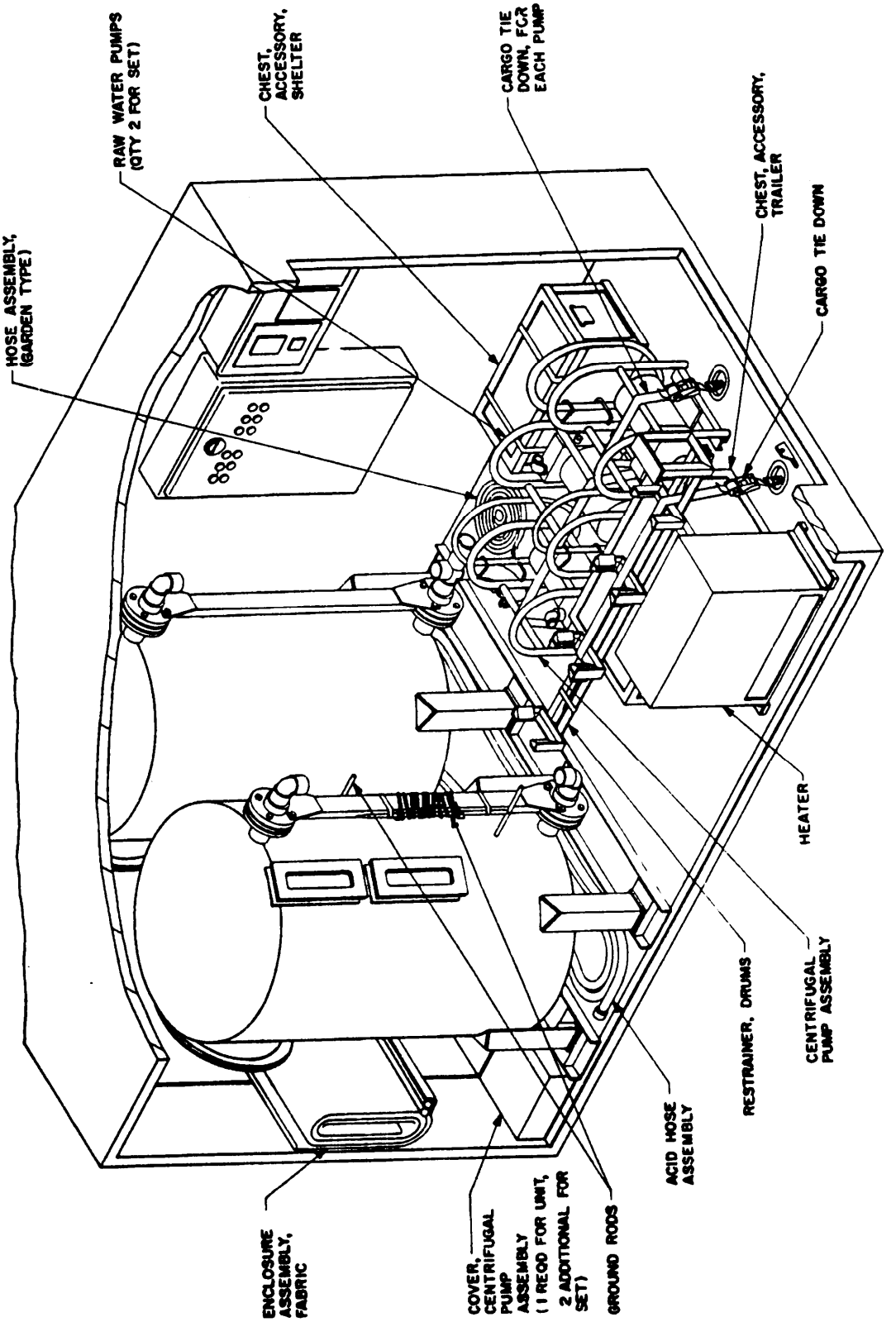
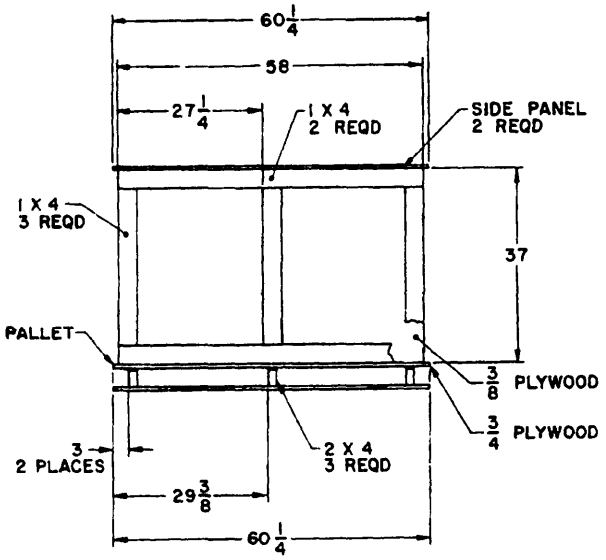
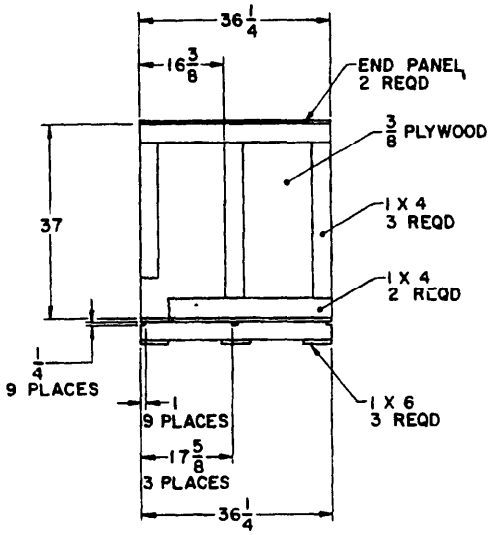


FIGURE 1. Location of specific unit and set components within shelter. X-2143A

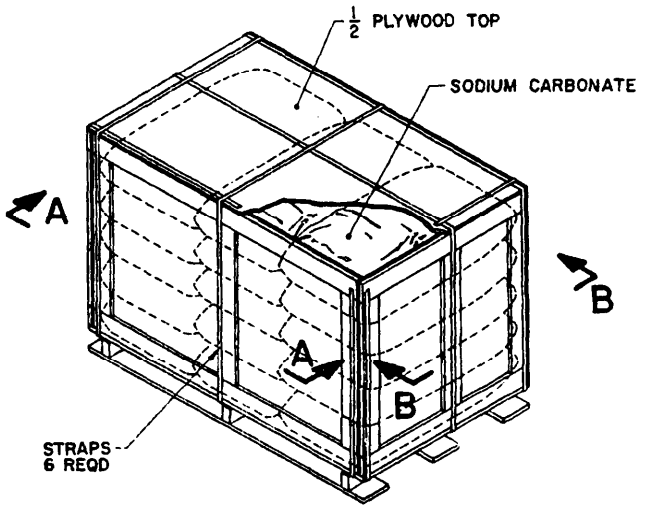
Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the currentversion before use.



VIEW **A-A**
ITEMS NOT SHOWN
OMITTED FOR CLARITY



VIEW **B-B**
STRAPS OMITTED
FOR CLARITY



NOTES:

1. PLYWOOD SHALL CONFORM TO NN-P-530, GROUP A; PS51, TYPE I, GRADE 3-4 OR GROUP B; PS1, GRADE C-D INTERIOR WITH EXTERIOR GLUE.
2. WOOD OTHER THAN PLYWOOD SHALL CONFORM TO MIL-STD-731, AND SIZES SHALL BE NOMINAL.
3. FABRICATE PALLET USING PALLET NAILS CONFORMING TO FF-N-105, TYPE II, STYLE 18, 2-1/2 MIN LONG, .120 DIA. CAUTION: DO NOT NAIL INTO CUTOUTS.
4. UNLESS OTHERWISE SPECIFIED HEREIN, FABRICATION OF SIDE AND END PANELS, AND TOP SHALL CONFORM TO PPP-B-601.
5. STRAPS SHALL BE IN ACCORDANCE WITH QQ-S-781, TYPE I OR IV, FINISH B, 3/4 X .028.
6. ENVELOPE DIMENSIONS MAY VARY DEPENDING ON CONTRACTOR'S BAG SIZE, AND MAY BE MODIFIED, PROVIDING MINIMUM CUBE IS REALIZED. CONSTRUCTION PRINCIPLES AND DETAIL REMAIN AS SHOWN (SEE 6.2).

FIGURE 2. Details for capped pallet for sodium carbonate.

X-2150A

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-D-52718A(ME) Demineralizer, Water, Ion Exchange Unit and Demineralization Equipment Set, Water, Ion Exchange 3,000 GPH:

NAME OF ORGANIZATION AND ADDRESS
Packaging of

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:10Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

MERADCOM
DRDME-DS
Ft. Belvoir, VA 22060

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research and
Development Command
DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.